

# Mike Pozulp

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Lawrence Livermore National Laboratory  
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United States

## Personal Information

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Citizenship: U.S.

## Interests

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Numerical Methods, Computer Architecture, Compilers

## Education

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### University of California, Berkeley

Berkeley, CA

May 2025

PhD, Applied Science & Technology

Thesis: "Hybrid Monte Carlo Deterministic Second Moment Method for Thermal Radiative Transfer"

Advisors: J. L. Vujic (UCB) and T. S. Haut (LLNL)

### The College of William & Mary

Williamsburg, VA

May 2015

Bachelor of Science, *magna cum laude*

Major: Computer Science

Minor: Economics

## Presentations and Publications

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Lead author indicated by \*

- **"A Hybrid Second Moment Method for Thermal Radiative Transfer"** (with T. Haut, P. Brantley, S. Olivier, J. Vujic). In *Proceedings of M&C 2025*. Denver, Colorado. April 2025.\*
- **"Status of Mercury and Imp: Two Monte Carlo Transport Codes Developed Using Shared Infrastructure at Lawrence Livermore National Laboratory"** (with B. Beck, R. Bleile, P. Brantley, S. Dawson, N. Gentile, E. Gonzalez, J. Grondalski, M. Lambert, M. McKinley, M. O'Brien, R. Procassini, D. Richards, A. Robinson, S. Sepke, D. Stevens, R. Vega, M. Yang). Published in *EPJ-N* and presented at *SNA+MC 2024*. Paris, France. October 2024.\*
- **"Achievement of Target Gain Larger than Unity in an Inertial Fusion Experiment"** (with O. Hurricane, A. Kritcher, A. Zylstra, D. Callahan, and many others). *Physics Review Letters* 132, 065102. February 2024.
- **"Noisy Radiation Diffusion in MFEM"** (with T. Haut, P. Brantley, J. Vujic). Presented at *MFEM Community Workshop*. Internet. October 2023.\*
- **"An Implicit Monte Carlo Acceleration Scheme"** (with T. Haut, P. Brantley, J. Vujic). In *Proceedings of M&C 2023*. Niagara Falls, Canada. August 2023.\*
- **"Progress Porting LLNL Monte Carlo Transport Codes to Nvidia GPUs"** (with R. Bleile, P. Brantley, S. Dawson, M. McKinley, M. O'Brien, A. Robinson, M. Yang). In *Proceedings of M&C 2023*. Niagara Falls, Canada. August 2023.\*
- **"Lawson Criterion for Ignition Exceeded in an Inertial Fusion Experiment"** (with O. Hurricane, A. Kritcher, A. Zylstra, D. Callahan, and many others). *Physics Review Letters* 129, 075001. August 2022.
- **"Fast Solvers for the Finite Element Method"** (with B. Muldoon). Unpublished. May 2022.
- **"Enhancements supporting IC usage of PEM libraries on next-gen platforms"** (with D. Richards, B. Ryujin). Technical Report LLNL-TR-823775, Lawrence Livermore National Laboratory, Livermore, California. June 2021.
- **"RISC-V Code Generation Comparison"** (with Y. Miyasaka). Unpublished. May 2021.\*
- **"Heterogeneity, Hyperparameters, and GPUs: Towards Useful Transport Calculations Using Neural Networks"** (with P. Brantley, T. Palmer, J. Vujic). In *Proceedings of M&C 2021*, 1252-1261. Raleigh, North Carolina. October 2021.\*

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|----------|----------------------------------------------------------------------------------------------------------------------|
| EPJ-N    | is the European Physics Journal - Nuclear Sciences & Technologies                                                    |
| SNA+MC   | is the Joint International Conference on Supercomputing in Nuclear Applications + Monte Carlo.                       |
| EuroLLVM | is the European LLVM Developers' Meeting.                                                                            |
| M&C      | is the International Conference on Mathematics and Computational Methods applied to Nuclear Science and Engineering. |
| SC       | is the International Conference for High Performance Computing, Networking, Storage, and Analysis.                   |

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|----------------------------------------|---------------|---------------------|
| <b>Lawrence Livermore National Lab</b> | Livermore, CA | July 2015 - Present |
| Position: Staff Scientist              |               |                     |

- ## Computer Skills

- ## Fellowships, Research Grants, and Contracts

- ## Honors, Prizes, and Awards

- ## Professional Development

- M&C 2025 Denver, Colorado 2025 April 27-30

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|-------------------------------------------|-------------------------|--------------------------|
| • SNA + MC 2024                           | Paris, France           | 2024 October 20-24       |
| • MFEM Community Workshop                 | Internet                | 2023 October 26          |
| • NECDC 2023                              | Los Alamos, New Mexico  | 2023 October 16-20       |
| • M&C 2023                                | Niagara Falls, Canada   | 2023 August 13-17        |
| • M&C 2021                                | Raleigh, North Carolina | 2021 October 3-7         |
| • J34 Applied Computer Science Meeting    | Livermore, California   | 2020 February 24-27      |
| • LLVM Developer Meeting                  | San Jose, California    | 2019 October 22-23       |
| • NSSC Fall Workshop                      | Livermore, California   | 2019 October 7-9         |
| • M&C 2019                                | Portland, Oregon        | 2019 August 25-29        |
| • LLVM Developer Meeting                  | San Jose, California    | 2018 October 17-18       |
| • J34 Applied Computer Science Meeting    | Albuquerque, New Mexico | 2018 February 11-16      |
| • Supercomputing (SC)                     | Denver, Colorado        | 2017 November 12-17      |
| • DoE CoE Performance Portability Meeting | Denver, Colorado        | 2017 August 21-24        |
| • Supercomputing (SC)                     | Salt Lake City, Utah    | 2016 November 13-18      |
| • DoE CoE Performance Portability Meeting | Glendale, Arizona       | 2016 April 18-22         |
| • ATPESC                                  | St. Charles, Illinois   | 2016 July 31 - August 12 |
| • Supercomputing (SC)                     | New Orleans, Louisiana  | 2014 November 16-21      |
| • Supercomputing (SC)                     | Denver, Colorado        | 2013 November 17-22      |

## Technical Coursework

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### University of California, Berkeley

|                                                            |             |
|------------------------------------------------------------|-------------|
| • Finite Elements in Nonlinear Continua (ME 280B)          | 2022 Spring |
| • Numerical Linear Algebra (MATH 221)                      | 2022 Spring |
| • Introduction to the Finite Element Method (ME 280A)      | 2021 Fall   |
| • Radiation Processes in Astronomy (PHY C207)              | 2021 Fall   |
| • Graduate Computer Architecture (CS 252A)                 | 2021 Spring |
| • Numerical Solution of Differential Equations (MATH 228B) | 2021 Spring |
| • Numerical Analysis (MATH 128A)                           | 2020 Fall   |
| • Nuclear Reactor Theory (NE 250)                          | 2020 Fall   |
| • Numerical Simulation in Radiation Transport (NE 255)     | 2018 Fall   |

### University of California, Davis

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|----------------------------------------------------------------|-------------|
| • Network Architecture & Resource Management (EEC 273/ECS 258) | 2018 Fall   |
| • Quantum Mechanics (PHY 115A)                                 | 2017 Spring |
| • Analytical Mechanics II (PHY 105B)                           | 2017 Winter |
| • Analytical Mechanics I (PHY 105A)                            | 2016 Fall   |

### University of California, San Diego

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|-----------------------------------------|-----------|
| • High Energy Density Physics (MAE 207) | 2017 Fall |
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### Stanford University

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|-----------------------------------------------------------|-------------|
| • Partial Differential Equations in Engineering (CME 204) | 2018 Winter |
| • Compilers (CS 143)                                      | 2016 Spring |
| • Introduction to Computer Graphics (CS 148)              | 2015 Fall   |

### The College of William & Mary

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|----------------------------------------------|-------------|
| • Random Walks in Biology (APSC 456)         | 2015 Spring |
| • Reliability (CS 668)                       | 2015 Spring |
| • General Physics II, Honors (PHYS 102H)     | 2015 Spring |
| • Analog Electronics (PHYS 252)              | 2015 Spring |
| • Ordinary Differential Equations (MATH 302) | 2014 Fall   |
| • General Physics I, Honors (PHYS 101H)      | 2014 Fall   |
| • Digital Electronics (PHYS 351)             | 2014 Fall   |
| • Finite Automata (CS423)                    | 2013 Fall   |
| • Operating Systems (CS 424)                 | 2013 Fall   |
| • Applied Financial Derivatives (ECON 415)   | 2013 Fall   |
| • Probability (MATH 401)                     | 2013 Fall   |
| • Numerical Analysis (MATH 413)              | 2013 Fall   |

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|------------------------------------------|-------------|
| • Programming Languages (CS 312)         | 2013 Spring |
| • Systems Programming (CS 415)           | 2013 Spring |
| • Econometrics (ECON 308)                | 2013 Spring |
| • Multivariable Calculus (MATH 212)      | 2013 Spring |
| • Algorithms (CS 303)                    | 2012 Fall   |
| • Computer Organization (CS 304)         | 2012 Fall   |
| • Intermediate Microeconomics (ECON 303) | 2012 Fall   |
| • Software Development (CS 301)          | 2012 Spring |
| • Database Systems (CS 321)              | 2012 Spring |
| • Intermediate Macroeconomics (ECON 304) | 2012 Spring |
| • Linear Algebra (MATH 211)              | 2012 Spring |
| • Data Structures (CS 241)               | 2012 Fall   |
| • Discrete Structures (CS 243)           | 2012 Fall   |